Description

Image





Caption

Thermosetting polyester used as the matrix for fibre reinforced products. © ANSYS, Inc

The material

Polyesters can be a thermosets, a thermoplastics or elastomers. The unsaturated polyester resins are thermosets. Most polyester thermosets are used in glass fiber/polyester composites. They are less stiff and strong than epoxies, but they are considerably cheaper.

Compositional summary

(OOC-C6H4-COO-C6H10)n

General properties

Density		1.04e3	-	1.4e3	kg/m^3
Price	*	1.55	-	1.97	GBP/kg
Date first used		1942			

Mechanical properties

Young's modulus		2.07	-	4.41	GPa
Shear modulus	*	0.744	-	1.59	GPa
Bulk modulus	*	4.49	-	4.71	GPa
Poisson's ratio		0.381	-	0.403	
Yield strength (elastic limit)	*	33	-	40	MPa
Tensile strength		41.4	-	89.6	MPa
Compressive strength	*	140	-	207	MPa
Elongation		2	-	2.6	% strain
Hardness - Vickers	*	10	-	22	HV
Fatigue strength at 10^7 cycles	*	16.6	-	35.8	MPa

Fracture toughness	*	1.09	-	1.69	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	0.00907	-	0.0193	

Thermal properties

Melting point	*	228	-	268	°C
Glass temperature	*	150	-	210	°C
Maximum service temperature		112	-	128	°C
Minimum service temperature	*	-43.2	-	6.85	°C
Thermal conductor or insulator?		Good insulator			
Thermal conductivity	*	0.287	-	0.299	W/m.°C
Specific heat capacity	*	1.51e3	-	1.57e3	J/kg.°C
Thermal expansion coefficient		99	-	180	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Good insulator			
Electrical resistivity		3.3e18	-	3e19	µohm.cm
Dielectric constant (relative permittivity)		2.8	-	4.4	
Dissipation factor (dielectric loss tangent)		0.003	-	0.04	
Dielectric strength (dielectric breakdown)		15	-	19.7	MV/m

Optical properties

Transparency		Transparent			
Refractive index		1.54	-	1.57	

Critical Materials Risk

High critical material risk?		No			

Processability

Castability		3	-	4	
Moldability		3	-	4	
Machinability		3	-	4	
Weldability		1			

Durability: water and aqueous solutions

Water (fresh)		Excellent			
Water (salt)		Excellent			
Soils, acidic (peat)		Limited use			
Soils, alkaline (clay)		Unacceptable			
Wine		Acceptable			

Durability: acids

Acetic acid (10%)	Limited use
Acetic acid (glacial)	Unacceptable
Citric acid (10%)	Excellent
Hydrochloric acid (10%)	Excellent
Hydrochloric acid (36%)	Excellent
Hydrofluoric acid (40%)	Unacceptable
Nitric acid (10%)	Excellent
Nitric acid (70%)	Unacceptable
Phosphoric acid (10%)	Excellent
Phosphoric acid (85%)	Excellent
Sulfuric acid (10%)	Excellent
Sulfuric acid (70%)	Excellent

Durability: alkalis

Sodium hydroxide (10%)	Unacceptable
Sodium hydroxide (60%)	Unacceptable

Durability: fuels, oils and solvents

Amyl acetate	Limited use
Benzene	Unacceptable
Carbon tetrachloride	Excellent
Chloroform	Unacceptable
Crude oil	Excellent
Diesel oil	Excellent
Lubricating oil	Excellent
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Excellent
Silicone fluids	Excellent
Toluene	Limited use
Turpentine	Excellent
Vegetable oils (general)	Acceptable
White spirit	Limited use

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Unacceptable
Acetone	Unacceptable
Ethyl alcohol (ethanol)	Unacceptable
Ethylene glycol	

	Excellent
Formaldehyde (40%)	Excellent
Glycerol	Excellent
Methyl alcohol (methanol)	Excellent

Durability: halogens and gases

Chlorine gas (dry)	Excellent
Fluorine (gas)	Unacceptable
O2 (oxygen gas)	Unacceptable
Sulfur dioxide (gas)	Excellent

Durability: built environments

Industrial atmosphere	Excellent
Rural atmosphere	Excellent
Marine atmosphere	Excellent
UV radiation (sunlight)	Fair

Durability: flammability

Flammability	Highly flammable

Durability: thermal environments

Tolerance to cryogenic temperatures	Unacceptable
Tolerance up to 150 C (302 F)	Excellent
Tolerance up to 250 C (482 F)	Unacceptable
Tolerance up to 450 C (842 F)	Unacceptable
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Geo-economic data for principal component

Annual world production, principal component	3.8e7	-	4.2e7	tonne/yr
Reserves, principal component	9.5e8	-	1.05e9	tonne

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 4.31	-	4.76	kg/kg
Embodied energy (virgin grade production)	* 93.5	-	103	MJ/kg
Water usage	* 190	-	210	l/kg

Material processing: energy

Polymer molding (injection, blow), embodied energy	* 27.1	-	32.9	MJ/kg
Machining, coarse, embodied energy	* 22.5	-	25.6	MJ/kg

Machining, fine, embodied energy	* 38.2	-	44.7	MJ/kg
Machining, grinding, embodied energy	* 53.9	-	63.7	MJ/kg

Material processing: climate change

Polymer molding (injection, blow), climate change (CO2-eq)	* 1.96	-	2.41	kg/kg
Machining, coarse, climate change (CO2-eq)	* 1.47	-	1.68	kg/kg
Machining, fine, climate change (CO2-eq)	* 2.5	-	2.93	kg/kg
Machining, grinding, climate change (CO2-eq)	* 3.53	-	4.17	kg/kg

Recycling and end of life

Recycle	✗			
Functional recycle	✗			
Combust for energy recovery	✓			
Heat of combustion (net)	* 28	-	29.4	MJ/kg
Combustion CO2	* 2.49	-	2.62	kg/kg
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

Thermosetting polyesters cannot be recycled.

Supporting information

Design guidelines

Thermosetting polyesters are the cheapest resins for making glass or carbon fiber composites, but they have lower strength than epoxies. They can be formulated to cure at or above room temperature. Modifications can improve the chemical resistance, UV resistance and heat resistance without too much change in the ease of processing. Polyester elastomers have relatively high moduli and are stronger than polyurethanes. They have good melt flow properties, low shrinkage, good resistance to oils and fuels. Polyester can be made conductive by adding 30% carbon fiber. As a tape, Mylar is used for magnetic sound recording. Unfilled polyester thermosetting resins are normally used as surface coatings but they tend to be brittle. of Thermosetting polyester has a corroding influence on copper

Technical notes

Polyesters are made by a condensation reaction of an alcohol like ethyl alcohol (the one in beer) and an organic acid like acetic acid (the one in vinegar). The two react, releasing water, and forming an ester.

Typical uses

Laminated structures, surface gel coatings, liquid castings, furniture products, bowling balls, simulated marble, sewer pipe gaskets, pistol grips, television tube implosion barriers, boats, truck cabs, concrete forms, lamp housings, skylights, fishing rods.

Tradenames

Celanex, Eastar, Hytrel, Plenco, Rynite, Synolite, Valox, Vybrex

Links

ProcessUniverse

Producers

Reference

Values marked * are estimates.
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